

Supporting Information for

Where predictive information resides in endogenous A-to-I RNA editing

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Supporting Information Text

This appendix documents the search over biological information scales and the evidence audit used to interpret it. The search separates intrinsic substrate properties, transcript and cellular realization, target-local selection, and statistical dependence among sites from one duplex. It includes branches that were selected, supported but not actionable, limited, unsupported in the tested formulation, deferred for unavailable matched data, or superseded by stronger lower-complexity evidence. Four evaluation regimes remain distinct: *development OOF* denotes pair-disjoint cross-fitted comparisons used for discovery and selection; *selection diagnostic* denotes published-validation quantities used for checkpoints or thresholds; *canonical held-out* denotes the single evaluation after the final manifest was frozen; and *secondary* denotes robustness or descriptive analyses that do not define the benchmark claim. These labels keep a negative result for an available measurement separate from a biological question that the benchmark could not validly test.

A. Benchmark reproduction and leakage control

Immutable AdarEdit benchmark. The upstream AdarEdit reproducibility release was preserved at the immutable revision recorded in repository provenance. Environment, package, GPU, input-file, and checkpoint provenance are retained in the repository. The upstream verifier recovered 884 graph-buildable inverted-Alu pairs: 566 train, 139 validation, and 179 test, with no pair shared between partitions. Supplied Baseline GAT checkpoints reproduced the reported metrics for artery tibial, brain/cerebellum, liver, skeletal muscle, and Combined. Combined pools evidence across GTEx tissues and was never treated as a fifth tissue when constructing environmental covariates.

Each example contains both Alu arms in transcript orientation, a focal adenosine, predicted dot-bracket structure, and EditingIndex-derived label. EditingIndex $\geq 15\%$ defines edited sites, EditingIndex $< 1\%$ defines nonedited sites, and the canonical binary task excludes the intermediate interval. Nodes encode nucleotide and structure-derived features; edges represent backbone and predicted base-pair relations. The original construction used ViennaRNA for secondary structure (1).

Coordinates and authoritative observations. Genomic coordinates were reconstructed with the original strand-aware transformation from local model positions to *hg38*/GRCh38. Pair intervals came from `Pair_Alu_withStrand.bed`; raw tissue tables supplied focal positions and EditingIndex. Both `site_id = chromosome:position:strand` and `(pair_id, local_position)` were retained. Quality control required an adenosine in transcript orientation, deterministic row mapping, correct strand, and stable identity across tissues. The resulting `site_master.parquet` records labels, continuous editing values, sequence, structure, coordinates, folds, OOF predictions, and provenance.

Test quarantine, grouped folds, and inference. Canonical test paths and SHA-256 values were registered before development. Supplied-checkpoint reproduction permitted aggregate metrics; candidate evaluation required authorization bound to the frozen manifest. Discovery used the published train and validation pairs divided deterministically into 5 outer folds by pair ID. In direct neural comparisons, training and preprocessing excluded both the outer fold and a separate internal-validation fold, which selected checkpoints and thresholds. Each neural OOF prediction therefore excluded its pair from training and selection.

The early RBP, genomic, chromatin, native-context folding, target-geometry, and initial substrate-profile residual screens used pooled `baseline_oof_logit` values. Their second-stage fitting and selection excluded the outer pair fold, but first-stage models generating second-stage training scores could include that fold. Consequently these are exploratory pooled-score screens, not fully nested end-to-end evaluations. Their shared-reference contrasts do not remove this dependency. Later hierarchical gates, stronger substrate-profile gates, and foundation fusion used explicitly outer-isolated first-stage fits. Pair-level geometry and matched-site tissue models used outcome-independent inputs and did not reuse the pooled baseline logits. Fully nested replication of the early screens remains necessary before treating their null increments as independently validated measurement limits.

The grouped OOF reference is an internal research instrument, not a revision of the published benchmark. Each outer model sees fewer pairs than a final fit and therefore has lower absolute performance. OOF comparisons estimate incremental value because candidate and reference share folds, training scope, and evaluation logic. The supplied-checkpoint canonical comparison answers the final benchmark question (Fig. S1). Development differences are never added arithmetically to published test scores.

We report AUROC, AUPRC, F1, precision, recall, specificity, balanced accuracy, Matthews correlation coefficient, and Brier score. Unless specified otherwise, thresholds maximized F1 on fold-internal validation data. Bootstraps sample pair IDs with replacement and retain every site from each sampled pair. Development comparisons used 1,000 or 2,000 replicates as recorded in their run summaries; within-duplex diagnostics used 5,000, tissue-variation and stability analyses used 2,000, and final-test comparisons used 2,000. Intervals condition on retained fitted predictions, without refitting or accounting for adaptive selection, and development intervals are not multiplicity-adjusted confirmatory tests. Cross-tissue repetition uses overlapping substrates, not independent cohorts.

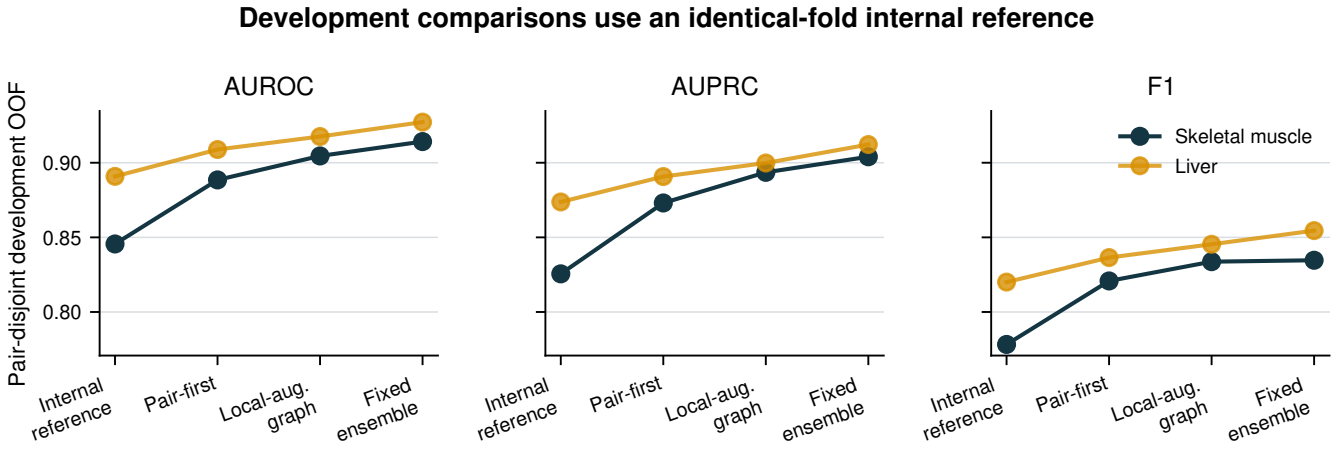


Fig. S1. Controlled neural comparisons in development. Candidate and internal reference predictions use identical pair folds. Absolute OOF scores are not substitutes for the separate canonical held-out comparison.

Table S1. Complete one-use canonical held-out metrics. Thresholds were frozen from the published validation partition before test access. Combined is a pooled endpoint.

Context	Model	AUROC	AUPRC	F1	Precision	Recall	Specificity	Brier
Artery	AdarEdit	0.9238	0.9262	0.8671	0.8345	0.9023	0.7930	0.1077
Artery	Selected predictor	0.9477	0.9485	0.8869	0.8840	0.8899	0.8650	0.0879
Brain	AdarEdit	0.9301	0.9286	0.8689	0.8371	0.9032	0.8021	0.1050
Brain	Selected predictor	0.9508	0.9504	0.8902	0.8741	0.9069	0.8529	0.0855
Liver	AdarEdit	0.9092	0.9242	0.8541	0.8368	0.8721	0.7788	0.1196
Liver	Selected predictor	0.9414	0.9500	0.8812	0.8946	0.8683	0.8670	0.0951
Skeletal muscle	AdarEdit	0.8688	0.8757	0.8139	0.7767	0.8549	0.7152	0.1526
Skeletal muscle	Selected predictor	0.9144	0.9207	0.8541	0.8230	0.8876	0.7788	0.1296
Combined	AdarEdit	0.9331	0.9356	0.8674	0.8334	0.9043	0.8026	0.1043
Combined	Selected predictor	0.9491	0.9513	0.8852	0.8463	0.9279	0.8159	0.0870

B. Baseline scope and transfer boundaries

Retained benchmark-characterization runs place the subsequent residual tests in context. The compact sequence–structure graph was evaluated within five human endpoints—four individual tissues and pooled Combined—and three nonhuman datasets. Within-context AUROC ranged from 0.869 to 0.933 for the human endpoints and from 0.823 to 0.865 for the nonhuman datasets. These values demonstrate substantial predictive information in sequence and predicted structure, but they are descriptive across datasets with different lineages, substrate classes, and data-generation procedures.

The full eight-by-eight transfer matrices train in one context and evaluate in every other context. AUROC is emphasized because it is not tied to transfer of a training-context decision threshold. Transfer among human endpoints was generally strong. Transfer among the three nonhuman datasets was more variable, and transfer between human and nonhuman settings was lower and asymmetric. The matrices therefore support sequence–structure regularities that transfer across contexts but not universally; they do not identify whether lineage, substrate class, assay, label distribution, or cellular environment causes the transfer loss. Exact values and their source paths are retained in `results/matrix_baseline_auroc.csv` and the other metric-specific matrices.

The accompanying within-context table compares the compact baseline with a Bio-aware variant. That variant changes both features and architecture and does not improve every endpoint. It is descriptive characterization, not a capacity-matched experiment or evidence that greater graph capacity alone fails. A separate graph-transformer/GATv2 direction was considered but not pursued after the target-local result.

The upstream study also reported strong GTEx–RNAAtlas concordance for shared editing measurements, supporting label reproducibility across cohorts and processing pipelines (2). The selected predictor was not evaluated on RNAAtlas; this observation therefore does not test transportability to an independently generated human cohort.

C. Search-space definition and evidence classes

The biological decomposition organizes an adaptive campaign, not one prospectively registered factorial search. Intrinsic-substrate tests examined whole-duplex sequence, predicted structure, native-context folding, and genomic organization. Context tests

examined same-site tissue variation and RBP, abundance, transcript, repeat, and chromatin measurements. Target-local tests asked what whole-duplex pooling leaves unused. The statistical branch separated label recovery from within-class pair weighting. Four model-sufficiency classes covered richer supervision, cross-context sharing, richer representations, and capacity or integration. Priorities reflected biological rationale, novelty relative to AdarEdit, data compatibility, leakage risk, and effective sample size; recorded outcomes determined subsequent branches.

A valid branch advanced only after comparison with its same-fold reference and relevant controls. Table S2 gives quantitative effects; the estimands differ and are not ranked on one common scale. Table S10 maps every retained direction to its scale, question, measurement or representation, comparator, evidence status, result, and disposition. Deferred donor-, cell-, regulator-, and in vivo structure measurements lacked benchmark-compatible outcomes and are not counted as negative evidence. The machine-readable record retains the historical priorities, sources, stopping criteria, and run-level artifacts.

D. Biological-context tests

RBP occupancy, binding potential, and abundance. Candidate RBPs were prioritized independently from model outcome using known ADAR interaction, dsRNA binding, perturbation evidence, nuclear localization, Alu association, expression, and assay availability. Experimental occupancy used replicated or high-confidence ENCODE eCLIP peaks (3); predicted occupancy used PrismNet and RBPNet where compatible (4, 5). Features included focal overlap, distance to the nearest peak, local-window signal, duplex-wide burden, arm asymmetry, partner occupancy, and regulator-class aggregates. GTEx medians supplied tissue abundance and binding-by-abundance interactions (6). Combined received no artificial tissue vector.

Controls included randomized RBP identities, shuffled tissue vectors, matched-dimensional noise, abundance without binding, and binding without abundance. High-confidence baseline false positives were matched on baseline probability, tissue, structural state, and pair characteristics. The pooled-score residual screens yielded no consistent increment beyond these controls (Fig. S2; section A). Repeat mapping further limits eCLIP sensitivity in Alu sequence, so peak absence was never interpreted as biological absence.

Label-switching sites hold genomic identity and the represented sequence and predicted structure fixed across tissues. Tissue identity nearly separated switch direction by itself. A separate continuous-difference analysis retained intermediate observations and compared RBP features with a tissue-pair mean reference under pair-disjoint fitting. Neither analysis recovered an RBP-specific increment. Unlike the residual screens, these comparisons do not reuse learned first-stage OOF logits. Matched cell- or donor-level occupancy and activity measurements would provide a more direct test of established RBP regulation (7–10).

Transcript, repeat, and chromatin annotations. Reconstructed pair intervals supplied arm separation and length, orientation, repeat subfamily and divergence, Alu-neighborhood composition, transcript overlap, and exon/intron annotations. The pooled-score residual screen combined standardized features with the baseline logit and found no consistent site-level increment (section A). Section E examines geometry at the pair level without this pooled-score dependency.

Exact-tissue compatible H3K36me3 tracks were summarized at the target, local windows, arms, and entire pair. A chromosome-shifted overlap acted as a spatial control. The muscle pair-span screen improved thresholded metrics without ranking gains; liver showed a small AUPRC effect without stable AUROC or F1 gains (Fig. S2). Neither pattern translated into consistent pair-level improvement. H2A.Z was deferred because adequately matched tissue data were unavailable. Quantitative, sample-matched assays remain a distinct test of cotranscriptional regulation (11, 12).

Native-gap/flank-aware RNAfold ensemble summaries were also screened against the pooled-score reference. Thermodynamic and inter-arm-pairing summaries gave mixed fold directions and no stable AUROC/F1 increment. This completed exploratory branch is distinct from deferred in vivo structure measurements (Table S10; report 12).

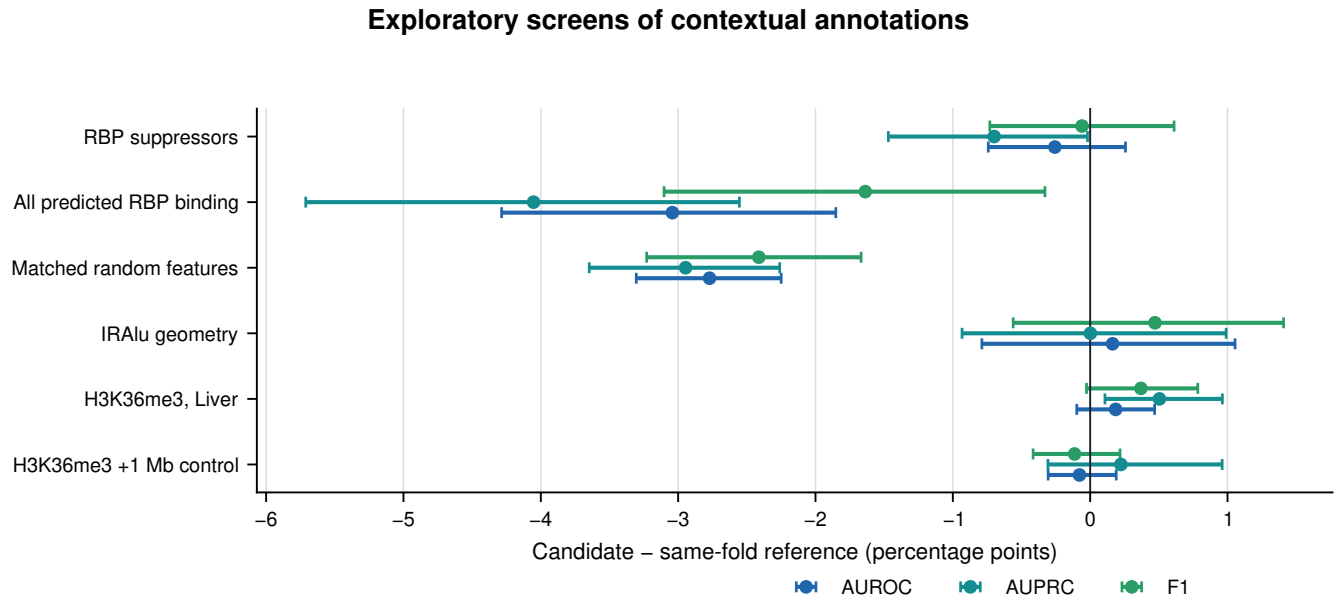


Fig. S2. Exploratory residual screens of biological annotations. Pair-clustered differences use the corresponding pooled-score reference. These second-stage screens are pair-disjoint but not fully nested with first-stage fitting (section A). The tested formulations did not consistently improve ranking and thresholded metrics.

E. Substrate-level editing burden and propensity

Pair-level analyses used one row per development substrate. Outcomes were edited-site fraction, mean EditingIndex, and whether a pair contained any high-editing site. Predictors ranged from adenosine count and graph-derived sequence/structure summaries to genomic geometry and the combined 16-feature low-capacity model. All fitting remained inside pair folds.

Inter-arm gap correlated negatively with edited-site fraction in every individual tissue. Geometry plus graph summaries improved R^2 over graph summaries alone in all four tissues, with positive pair-clustered intervals (Table S3). The combined model also classified whether a pair contained any high-editing site above chance. These observations establish reproducible substrate-level variation in editing burden. They are consistent with a realization or engagement dimension, but neither molecular process was measured directly.

Site-level translation was weaker. The outer-isolated burden gate modestly improved AUPRC over expanded row-wise training, but its F1 interval included zero and Brier score worsened. A separate neural auxiliary burden objective was compared with the stronger pair-first model and met its stopping criterion. Thus the tested integrations did not yield a stable site-level correction. Neither identifies the biological source of residual error.

Table S2. Development evidence synthesis. Each row states its comparator; estimands differ across rows. Values are percentage-point medians with 95% pair-clustered intervals unless noted. RBP, transcript/repeat, chromatin, and target/contact residual screens reuse pooled first-stage OOF scores and are exploratory, not fully nested (section A). "Not supported" applies to the tested formulation and selection criterion.

Axis	Hypothesis	Comparator	Primary effect [95% CI]	Secondary behavior	Evidence status	Interpretation
Supervision	Additional valid high/low labels	Original internal reference; same folds	ΔAUROC +2.04 [+1.06, +3.00]	ΔF1 +1.42 [+0.21, +2.57]	Selected	Additional valid labels contribute independently.
Statistical unit	Pair-first weighting	Expanded row-wise sampling; identical labels	ΔAUROC +2.27 [+1.25, +3.36]	ΔF1 +2.85 [+1.68, +4.01]	Selected	Correcting substrate weight adds a second gain.
Supervision	Continuous + ordinal auxiliary labels	Pair-first binary model	ΔAUROC +0.81 [+0.10, +1.51]	ΔF1 +0.08 [-0.71, +0.86]	Limited	Ranking/calibration gain; negligible F1; adverse natural-prevalence integration.
Representation	Explicit focal-node retention	Pair-first whole-duplex pooling	ΔAUROC +1.26 [+0.52, +2.03]	ΔF1 +0.43 [-0.59, +1.45]	Limited	Ranking gain; standalone F1 interval crosses 0.
Representation	Target/contact geometry	Pooled-score reference (exploratory)	ΔAUROC +1.05 [+0.46, +1.61]	ΔF1 +0.19 [-0.66, +1.09]	Limited	Ranking and calibration gains localize the target effect.
Representation	Aligned target/helix-partner representation	Pair-first whole-duplex-only predictor	ΔAUROC +1.61 [+0.80, +2.38]	ΔF1 +1.31 [+0.27, +2.30]	Selected	Target-aligned fusion improves all three metrics.
Capacity control	96-nt shifted local window	Pair-first whole-duplex-only predictor	ΔAUROC -17.65 [-19.69, -15.70]	ΔF1 -11.80 [-13.37, -10.36]	Not supported	Matched capacity without target alignment is strongly adverse.
Integration	Fixed equal-logit integration	Whole-duplex-only predictor; muscle	ΔAUROC +2.56 [+2.07, +3.05]	ΔF1 +1.39 [+0.45, +2.34]	Selected	Complementary predictions; replicated in liver.
Substrate propensity	Pair geometry and editing burden	Geometry + graph summaries vs graph summaries	ΔR^2 +4.12 to +21.64 points (4 tissues)	All four pair-clustered 95% intervals exclude 0	Supported, not yet actionable	Burden is predictable; consistent with variation in realization/engagement.
Biology × AI	Cross-fitted engagement gate	Expanded row-wise site predictor	ΔAUPRC +0.84 [+0.19, +1.47]	ΔF1 +0.41 [-0.14, +0.95]	Limited	AUPRC gain; F1 interval crosses 0; Brier worsens.
Transfer	Shared four-tissue learning	Independent tissue models; mean across 4 tissues	ΔAUROC +0.87 [+0.58, +1.15]	ΔF1 +0.75 [+0.37, +1.13]	Limited	Positive transfer below preset criteria; mean Brier worsens.
Biological context	Bulk H3K36me3 occupancy	Pooled-score reference; liver (exploratory)	ΔAUPRC +0.50 [+0.11, +0.96]	ΔF1 +0.37 [-0.03, +0.78]	Limited	Liver: AUPRC only. Muscle threshold effects did not extend to ranking; no stable pair-level gain.
Biological context	RBP binding × bulk abundance	Pooled-score reference (exploratory)	ΔAUROC -0.26 [-0.74, +0.26]	ΔF1 -0.06 [-0.73, +0.61]	Not supported	No consistent screen gain; broader predicted panels were adverse.
Biological context	Transcript/repeat annotation panel	Pooled-score reference (exploratory)	ΔAUROC -0.41 [-1.69, +0.90]	ΔF1 -0.90 [-2.31, +0.38]	Not supported	No improvement beyond the graph representation.
Representation	Frozen RiNALMo target embedding	Recalibrated cross-fitted baseline	ΔAUROC -0.05 [-0.51, +0.39]	ΔF1 +0.70 [+0.06, +1.34]	Not supported	F1-only threshold effect; no AUROC/AUPRC gain; failed the preset rule.
Representation	Substrate-wide all-A decoder	Pair-first predictor on identical rows	ΔAUROC -6.64 [-9.25, -4.38]	ΔF1 -5.99 [-8.78, -3.65]	Not supported	Prespecified two-fold futility criterion met.
Integration	Unified end-to-end architecture	Immutable fixed ensemble	ΔAUROC -0.72 [-1.29, -0.15]	ΔF1 -0.30 [-1.25, +0.66]	Not supported	AUROC, AUPRC, and calibration were worse.
Descriptive comparison	Supplied Bio-aware variant	Supplied Bio-aware vs compact checkpoints	No common OOF effect estimate	Not uniformly better across endpoints	Mixed; descriptive	Features and architecture change together; capacity is not isolated.

Table S3. Pair geometry identifies a reproducible component of substrate-level editing burden. Spearman ρ relates genomic inter-arm gap to edited-site fraction. The low-capacity pair model combines genomic geometry with graph-derived summaries; ΔR^2 is relative to graph summaries alone with a 95% pair-clustered bootstrap interval. "Any high" denotes whether a pair contains at least one site with EditingIndex $\geq 15\%$. These outcomes do not directly measure molecular realization or engagement.

Context	Gap–burden ρ	Graph-only R^2	Combined R^2	ΔR^2 [95% CI], points	Any-high AUROC
Artery	-0.482	0.053	0.270	+21.64 [+16.39, +26.93]	0.816
Brain	-0.425	0.050	0.239	+19.11 [+13.94, +23.68]	0.749
Liver	-0.475	0.070	0.277	+20.82 [+15.75, +26.03]	0.728
Skeletal muscle	-0.334	0.010	0.050	+4.12 [+0.99, +6.98]	0.705

F. Richer supervision and the statistical unit

The expanded training pool contains all eligible high- and low-editing observations in the training pairs. Both formulations allocate equal numbers of draws to the two classes (up to one draw for an odd epoch budget). The row-wise formulation samples sites uniformly within each class. The pair-first formulation instead samples uniformly among pairs containing that class, then uniformly among the eligible sites in the chosen pair. Pairs lacking a class are absent from that class's sampling pool; there is no fallback draw. Pair weighting is therefore uniform within class, not necessarily across all pairs. The canonical-row epoch budget was fixed.

The same-fold sequence original reference → expanded row-wise → expanded pair-first separates the two corrections (Table S4). Label recovery improved all three primary metrics. With the label pool and class balance fixed, pair-first weighting added a second increment, including +2.85 F1 points. The combined expanded pair-first formulation improved every endpoint (Table S5); the weighting-only decomposition was conducted in muscle.

The binary/ordinal/continuous model shared an encoder across canonical classification, ordinal bins, and transformed EditingIndex regression. It added a small ranking and Brier improvement over pair-first binary training, but essentially no F1. Adding it to the selected predictor reduced natural-prevalence F1, so the graded-label branch remained limited. Intermediate values contain ordering information, but they did not improve the primary thresholded task enough to justify added complexity.

Table S4. Same-fold decomposition of expanded supervision and pair-first weighting in muscle development data. Intervals are pair-clustered bootstrap intervals for the incremental step indicated, in percentage points.

Metric	Original reference	All high/low, row-wise	Δ from labels [95% CI]	All high/low, pair-first	Δ from pair weighting [95% CI]
AUROC	0.8456	0.8659	+2.04 [+1.06, +3.00]	0.8885	+2.27 [+1.25, +3.36]
AUPRC	0.8256	0.8520	+2.63 [+1.25, +4.17]	0.8730	+2.08 [+0.76, +3.55]
F1	0.7783	0.7924	+1.42 [+0.21, +2.57]	0.8208	+2.85 [+1.68, +4.01]

Table S5. Pair-first expanded supervision versus the same-fold internal AdarEdit reference. Values are percentage-point median differences with 95% pair-clustered bootstrap intervals.

Context	ΔAUROC [95% CI]	ΔAUPRC [95% CI]	ΔF1 [95% CI]
Artery	+1.22 [+0.80, +1.66]	+1.37 [+0.79, +1.97]	+0.69 [+0.10, +1.27]
Brain	+1.17 [+0.82, +1.54]	+1.64 [+1.02, +2.20]	+1.11 [+0.61, +1.64]
Liver	+1.79 [+1.18, +2.37]	+1.72 [+0.70, +2.71]	+1.64 [+0.96, +2.30]
Skeletal muscle	+4.28 [+3.07, +5.49]	+4.73 [+3.15, +6.35]	+4.23 [+3.11, +5.38]
Combined	+1.45 [+1.09, +1.84]	+1.69 [+1.15, +2.24]	+1.41 [+0.94, +1.98]

G. Focal and target-local representation

Three stages tested whether whole-duplex mean pooling loses site identity. First, fold-isolated target-contact geometry added AUROC and AUPRC but little F1, localizing a ranking effect near the target. Second, a focal-node graph model preserved the final target embedding alongside the global mean. It improved ranking relative to the pair model and an otherwise matched duplicate-global control, although its standalone F1 interval was insufficient for selection. Third, the selected local-augmented graph model aligned a 65-position representation of the focal adenosine and its dot-bracket helix partner. Two dilated convolutional layers encoded nucleotide, pairing, validity, target/helix-partner indicators, and relative position; the result was fused with the whole-duplex graph embedding before classification.

The local-augmented graph model exceeded the whole-duplex-only predictor in muscle and replicated in liver. Its matched-capacity control moved the same local window 96 nt away while retaining the graph, dimensions, and optimization. The shifted model collapsed across AUROC, AUPRC, and F1 (Table S6), showing that the predictive gain depends on correct target centering rather than generic convolutional capacity. Because the full paired window was shifted, this control does not isolate the helix partner from the broader target-centered neighborhood. Natural-prevalence results preserved the advantage (Fig. S3).

Table S6. Target-representation controls in muscle development OOF data. All values are evaluated on the same canonical development rows.

Representation	AUROC	AUPRC	F1	Brier
Whole-duplex pair model	0.8885	0.8730	0.8208	0.1372
Explicit focal node	0.9014	0.8902	0.8251	0.1342
Graph + aligned target/helix-partner	0.9045	0.8936	0.8337	0.1286
Matched-capacity +96-nt shift	0.7122	0.6943	0.7028	0.2356

H. Cross-tissue sharing

A four-output masked model shared a graph encoder across individual tissues while omitting unobserved labels from the loss. Relative to independently trained pair-first models, it yielded limited positive transfer, concentrated in the data-limited skeletal-muscle endpoint. Mean AUROC/AUPRC/F1 changes across the four individual tissue endpoints were 0.0087/0.0114/0.0075, but the predeclared AUROC and F1 selection criteria were not both met and mean Brier increased by 0.0043. Transferable intrinsic rules therefore exist, but their effect was smaller and less calibrated than the improvements from supervision and target alignment. Combined was excluded from this biological multitissue formulation.

I. Foundation-model and substrate-wide experiments

Frozen RiNALMo representations were evaluated at the target and substrate scales, including a larger frozen variant (13). Residual models and neural fusion used fold-fitted normalization and matched-dimensional controls. The 150M target view reproduced a small F1-only threshold effect, but neither variant improved AUROC/AUPRC or met the predeclared requirement to beat matched random features on both ranking and F1. The more specific next question is whether task-specific pretraining on inverted-Alu substrates would help. A corpus audit found 288,056 benchmark-excluded opposite-orientation Alu pairs in GENCODE transcript spans, but transcript connectivity, foldability, and sequence-redundancy filters were insufficiently resolved for a leakage-safe test, so that direction remains deferred.

The substrate-wide decoder encoded a pair once and classified all candidate adenosines jointly. It was evaluated against a pair-first model on identical rows with a prespecified two-fold futility criterion. Decoder-minus-reference differences were -0.0664 AUROC and -0.0599 F1, with wholly adverse intervals, and training stopped. A relative within-pair rank feature helped a weaker original encoder in a pooled-score screen, but its later outer-isolated test neither beat pair-first training nor the shuffled-rank control. The tested joint decoder therefore did not improve on pair-first stochastic learning.

J. Natural-prevalence robustness

The canonical benchmark is balanced for direct comparison, so its precision is not a deployment-prevalence estimate. The secondary natural-prevalence evaluation retained every eligible high/low observation from unchanged development pairs. In both muscle and liver, the fixed ensemble improved ranking, AUPRC, F1, and false-positive count over its pair-first whole-duplex-only component (Fig. S3; Table S7). These results support practical false-positive reduction but do not replace the canonical task.

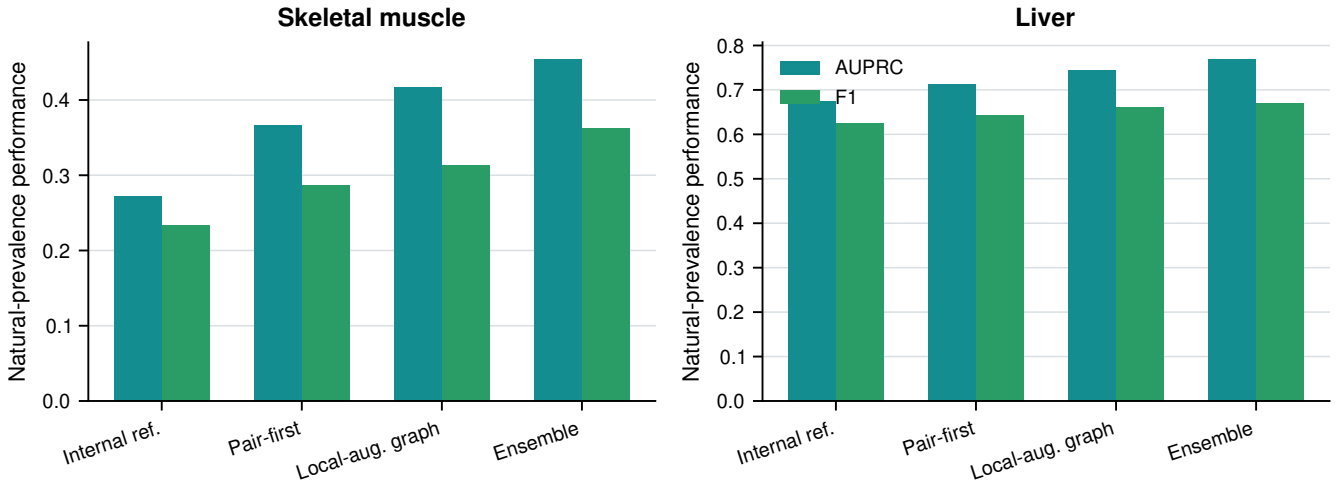


Fig. S3. Secondary natural-prevalence development evaluation. Performance for the same-fold internal reference, pair-first whole-duplex-only model, local-augmented graph model, and fixed ensemble on all eligible high/low observations from unchanged muscle and liver development pairs.

Table S7. Secondary natural-prevalence development evaluation. All eligible high/low observations from unchanged development pairs are included; fold-specific validation thresholds are used.

Context	Model	n	AUROC	AUPRC	F1	Precision	Recall	FP
Skeletal muscle	Internal reference	54289	0.8478	0.2711	0.2328	0.1351	0.8415	15804
Skeletal muscle	Pair-first whole-duplex	54289	0.8900	0.3659	0.2868	0.1714	0.8773	12441
Skeletal muscle	Local-augmented graph	54289	0.9072	0.4166	0.3125	0.1900	0.8800	11007
Skeletal muscle	Fixed ensemble	54289	0.9165	0.4540	0.3621	0.2306	0.8422	8245
Liver	Internal reference	38096	0.8915	0.6732	0.6246	0.4906	0.8592	6690
Liver	Pair-first whole-duplex	38096	0.9094	0.7128	0.6418	0.5056	0.8785	6442
Liver	Local-augmented graph	38096	0.9200	0.7432	0.6602	0.5265	0.8851	5969
Liver	Fixed ensemble	38096	0.9290	0.7687	0.6684	0.5306	0.9028	5988

K. Preregistered unified end-to-end experiment

After the prediction-level ensemble was frozen but before test access, a separate protocol specified one connected model. A shared three-layer graph attention encoder produced whole-duplex and focal-node states; the aligned local CNN supplied the third view. The three projected representations and their pairwise elementwise interactions entered a 64-unit fusion layer, while a training-only substrate head carried fixed loss weight 0.25. The connected model had 65,698 trainable parameters versus 80,138 parameters across the two independent ensemble components. Training used pair-first class balance, 1,000 epochs, Adam at learning rate 10^{-3} and weight decay 10^{-4}, batch size 256, and seed 20,260,821. The comparator prediction hash and staged decision rules were preregistered.

The unified model's AUROC and AUPRC were below the frozen ensemble in every outer fold (Fig. S4; Table S8). Aggregate F1 was similar, but ranking intervals were entirely adverse and calibration was substantially worse. The conditional no-auxiliary repair, liver replication, and all-context stages were therefore not opened. This preserves the decision rule and establishes that, in this comparison, independently trained representations were more reliably combined at prediction level than through the tested joint optimization. The experiment does not identify why optimization differed.

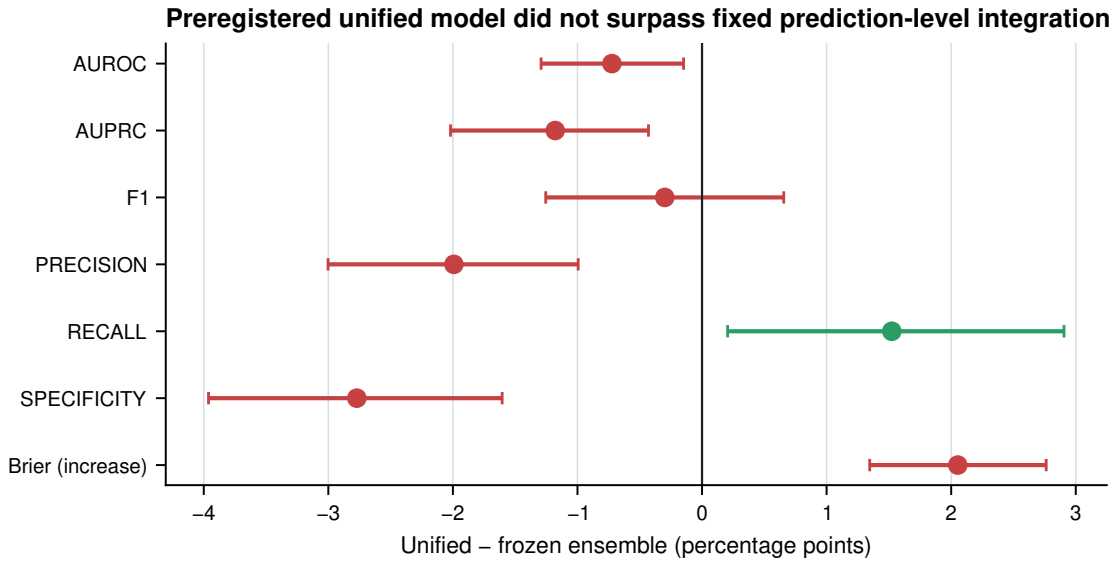


Fig. S4. Preregistered unified model versus the immutable ensemble. Bars are 95% pair-clustered bootstrap intervals for unified minus ensemble in muscle development OOF data. A positive Brier difference is adverse.

Table S8. Preregistered unified-model comparison on skeletal-muscle development data. The unified architecture was rejected before canonical-test access. Pair-clustered intervals compare the unified model with the immutable ensemble.

Model	AUROC	AUPRC	F1	Precision	Recall	Specificity	Brier
Internal reference	0.8456	0.8256	0.7783	0.7238	0.8415	0.6894	0.1695
Unified end-to-end	0.9072	0.8922	0.8317	0.8074	0.8575	0.8022	0.1372
Frozen ensemble	0.9142	0.9040	0.8347	0.8273	0.8422	0.8299	0.1168
Local-augmented graph	0.9045	0.8936	0.8337	0.7920	0.8800	0.7765	0.1286

Metric	Unified − ensemble, points [95% CI]
AUROC	-0.72 [-1.29, -0.15]
AUPRC	-1.18 [-2.02, -0.43]
F1	-0.30 [-1.25, +0.66]
PRECISION	-1.99 [-3.00, -0.99]
RECALL	+1.52 [+0.21, +2.91]
SPECIFICITY	-2.77 [-3.96, -1.60]
BRIER	+2.05 [+1.35, +2.76]

L. Final training and frozen canonical evaluation

For each endpoint, a pair-first whole-duplex-only graph model and an independently trained pair-first graph model augmented with the aligned target/helix-partner convolution were fitted on the published training pairs. Published validation data selected checkpoints and one ensemble threshold per endpoint. Ten checkpoints, histories, configurations, selected epochs, thresholds, inputs, code revision, and weights were written to `provenance/final_pipeline_manifest.json`, which retains the exact manifest SHA-256 and bound code revision. The canonical test was evaluated once after this freeze, and no subsequent model decision used test outcomes.

AUROC and AUPRC improved in every endpoint (Fig. S5; Table S1). At frozen thresholds, false-positive rescue rates exceeded true-positive damage rates. Matched-recall counts come from descriptive rethresholding of final predictions, with pair-clustered uncertainty (Table S9). Brier score decreased throughout; reliability curves show the remaining calibration deviations (Fig. S6). A lower Brier score combines changes in discrimination and calibration and does not isolate either.

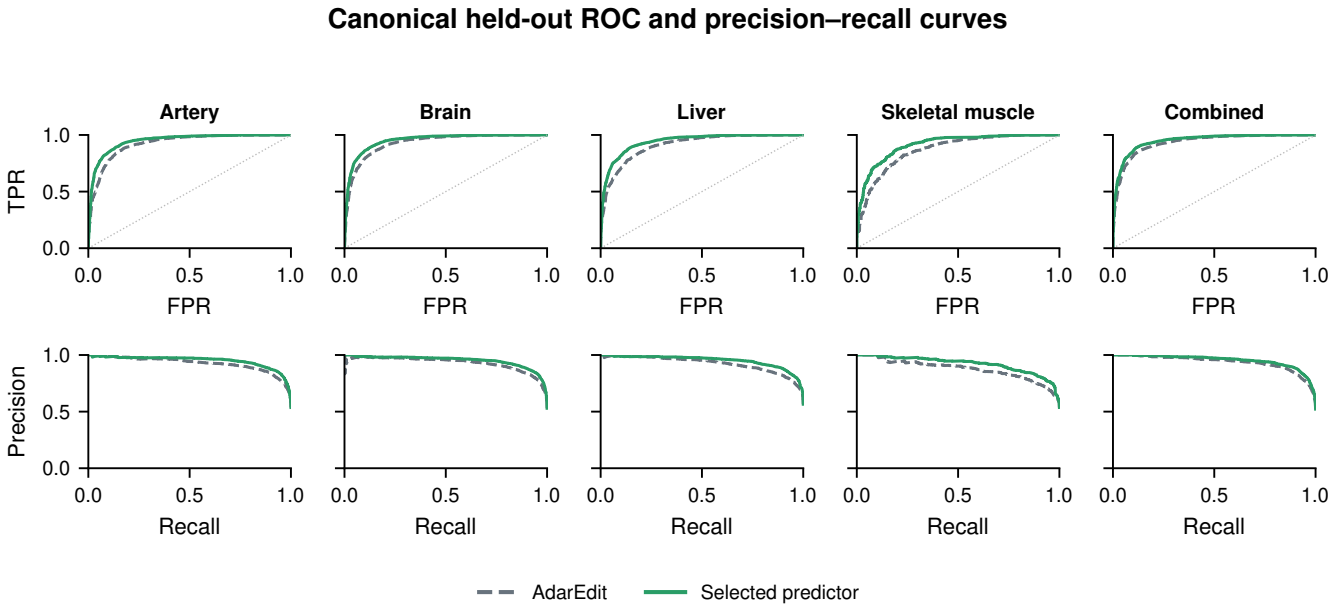


Fig. S5. Canonical held-out ROC and precision–recall curves. The selected predictor and supplied-checkpoint AdarEdit are shown for four individual tissue endpoints and Combined.

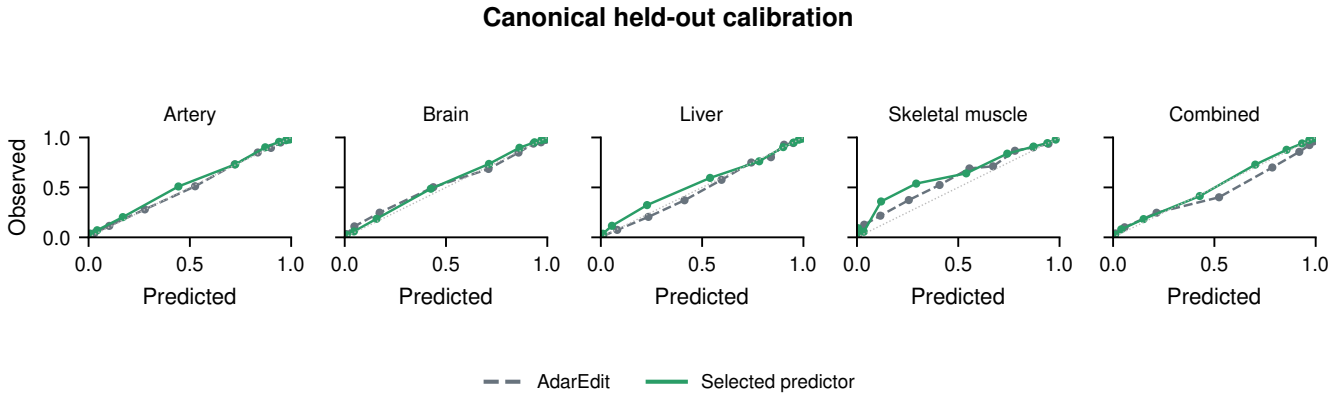


Fig. S6. Canonical held-out calibration. Reliability curves for the selected predictor and supplied-checkpoint AdarEdit. Brier score is lower for the selected predictor in every endpoint.

Table S9. Canonical-test false-positive transitions. Rescue and damage use frozen validation-selected thresholds. The final column is a post-test descriptive analysis that rethresholds the selected predictor to match published-baseline recall exactly.

Context	Baseline FP	FP corrected (%)	New FP	TP damaged (%)	FP avoided at matched recall
Artery	460	202 (43.9%)	42	113 (4.9%)	126
Brain	425	164 (38.6%)	55	73 (3.3%)	119
Liver	399	202 (50.6%)	43	121 (5.9%)	158
Skeletal muscle	188	83 (44.1%)	41	28 (4.3%)	68
Combined	459	147 (32.0%)	116	38 (1.7%)	148

M. Search-space coverage and provenance

Table S10 records search coverage and branch dispositions. Each run in `experiments/experiments.csv` records code revision, input hashes, folds, seed, features, hyperparameters, artifacts, metrics, and test access. OOF predictions, controls, ablations, and bootstrap tables remain in `experiments/runs/`; reports 00–29 preserve decisions and provenance. The numerical claim ledger maps manuscript values to authoritative artifacts and selectors.

Table S10. Search-space coverage and final disposition. Rows group the adaptive campaign by question and distinguish controlled, exploratory, descriptive, and untested evidence. "Deferred" means suitable data or a validated resource was unavailable, not that the hypothesis failed. Quantitative effects are in Table S2; historical priorities, stopping rules, and run artifacts remain in the machine-readable record.

Scale	Question	Candidate information or measurement	Comparator or control	Status and evidence regime	Principal outcome	Disposition
Intrinsic substrate	How much is recoverable from the duplex itself?	Whole-duplex sequence and predicted secondary structure	Compact AdarEdit sequence–structure graph; pair-disjoint benchmark	Supported reference *Retained benchmark characterization*	Substantial within-context signal; transfer is structured but not universal	Reference for all incremental tests
Intrinsic substrate	Does genomic organization add information absent from the folded graph?	Inter-arm gap, span, arm geometry, graph summaries, edited-site burden	Geometry plus graph summaries versus graph summaries alone; shuffled targets	Supported pair-level *Development pair-level*	Geometry adds reproducible between-substrate burden signal in all four tissues	Retain biological result; not a selected site feature
Intrinsic substrate	Do broad static annotations improve individual-site prediction?	Repeat subfamily/divergence/neighborhood and transcript/exon/intron annotations	Pooled-score residual reference; matched random block	Not supported in screen *Exploratory pooled-score screen*	No stable residual increment in the tested screen	Not selected; fully nested replication remains open
Intrinsic substrate	Do native sequence context and folding ensembles add signal?	Native-gap and flank-aware folding thermodynamics; inter-arm pairing summaries	Pooled-score reference; matched-dimensional random features	Limited *Exploratory pooled-score screen*	Mixed fold directions; no stable AUROC or F1 increment	Completed; distinct from untested in vivo conformation
Context / realization	Do fixed sites change across tissue contexts?	Matched-site EditingIndex, strict edited/nonedited switches, matched-site pair summaries	Same genomic sites across four individual tissues; Combined excluded	Supported descriptive *Post-hoc development observation*	Tissue-dependent site outcomes coexist with stable substrate rankings	Retain as positive evidence of context dependence
Context / realization	Do available RBP measurements resolve residual site variation?	ENCODE eCLIP, PrismNet, RBPNet, bulk GTEx abundance and interactions	Pooled-score reference; randomized RBP identities, shuffled tissue vectors, matched noise; tissue-effect reference	Not supported in tested comparisons *Exploratory residual screens; pair-disjoint matched-site models*	No consistent increment in residual screens or matched-site tissue models	Not selected; do not infer biological absence
Context / realization	Does bulk chromatin add substrate or site information?	Exact-tissue H3K36me3 peak overlap	Pooled-score reference; +1-Mb shifted-locus control; pair-level geometry reference	Limited *Exploratory site screens; pair-level development*	Muscle threshold and liver AUPRC effects; no consistent cross-metric or pair-level gain	Not selected; finer matched assays remain untested
Context / realization	Can other chromatin states be tested with matched coverage?	H2A.Z and other quantitative chromatin measurements	Tissue-match and coverage feasibility audit	Deferred *Feasibility only*	No adequately matched and powered tissue resource was available	New matched data required; not negative evidence
Context / realization	Can substrate propensity improve site prediction?	Cross-fitted burden gate and neural auxiliary engagement head	Gate versus expanded row-wise; auxiliary head versus pair-first; graph-only and shuffled-engagement controls	Limited *Outer-isolated development comparisons*	Pair burden is learnable but does not yield a stable site-level correction	Retain pair-level insight; do not select gate
Context / realization	Can matched cellular state be tested directly?	Donor/cell-resolved editing, ADAR isoform and IFN state, localization, regulator perturbation, in vivo structure	Resource compatibility, coverage, and leakage audit	Deferred *Feasibility only*	Required outcomes and state measurements are not matched to benchmark sites at adequate scale	New matched data required; not negative evidence
Target-local selection	Is target-specific information lost by whole-duplex pooling?	Target/contact geometry and explicit focal-node retention	Pooled-score geometry screen; same-fold neural reference and duplicate-global width control	Limited but informative *Exploratory geometry screen; controlled neural OOF*	Ranking gains localize additional signal near the target	Use to motivate aligned local representation
Target-local selection	Does correctly centered local context add information?	Target/helix-partner-aligned local representation	Pair-first whole-duplex predictor; matched-capacity window shifted by 96 nt in muscle	Selected *Neural OOF; local gain repeated in liver*	Aligned local context improves prediction; the shifted representation is strongly adverse	Retain selected local component; partner identity not isolated
Target-local selection	Do global and local views remain complementary?	Fixed 0.5/0.5 equal-logit combination	Whole-duplex-only predictor; no weight search	Selected *Development OOF with liver replication*	Independent whole-duplex and local-augmented views improve combined prediction	Retain frozen synthesis
Target-local selection	Does a site's predicted rank within its substrate add information?	All-adenosine relative-rank and broad substrate-profile features	Pooled-score original reference; later outer-isolated pair-first reference and within-pair shuffled rank	Limited / superseded *Exploratory profile screen followed by outer-isolated development gate*	Initial rank-screen gain did not survive the stronger outer-isolated comparison	Do not select relative-rank correction; distinguish from direct joint decoder
Statistical dependence	Do additional valid labels help before weighting changes?	Expanded high/low labels under row-wise sampling	Original internal reference; identical folds and fixed epoch budget	Selected *Development OOF in muscle*	Recovering valid labels adds a separable gain	Retain expanded training pool

Table continued

Scale	Question	Candidate information or measurement	Comparator or control	Status and evidence regime	Principal outcome	Disposition
Statistical dependence	Does correcting substrate weight add a second gain?	Class-balanced pair-first sampling on expanded labels	Expanded row-wise sampling; identical folds, class counts, labels, and epoch budget	Selected *Weighting ablation in muscle; combined formulation across five endpoints*	Pairs are weighted uniformly within each class before sites are drawn	Retain class-balanced pair-first learning
Model sufficiency	Does richer supervision or cross-tissue sharing add enough information?	Continuous/ordinal auxiliary tasks and masked four-tissue shared learning	Pair-first binary or independent-tissue predictors; Combined excluded	Limited *Development OOF*	Small ranking or transfer gains miss selection criteria and worsen calibration or prevalence behavior	Retain concise boundary; omit from selected predictor
Model sufficiency	Do frozen pretrained RNA representations recover more signal?	35M and 150M RiNALMo embeddings	Outer-isolated graph reference; matched random and shuffled features	Not supported for selection *Outer-isolated development OOF*	Small F1-only effect; no control-supported ranking gain	Do not select frozen fusion; other pretraining remains open
Model sufficiency	Does the supplied richer variant improve prediction?	Bio-aware features and architecture	Supplied Bio-aware versus compact checkpoints	Descriptive mixed result *Retained benchmark characterization*	No uniform advantage; features and architecture change together	Not an isolated test of graph capacity
Model sufficiency	Should a graph-transformer expansion be opened after the local result?	Graph Transformer / GATv2 multiscale encoder	Value-of-information review after the lower-capacity aligned-local gain	Superseded / not pursued *Protocol decision*	The stronger targeted result removed the rationale for a larger generic search	Do not interpret as negative evidence
Model sufficiency	Can task-specific pretraining be tested without leakage?	Self-supervised inverted-Alu pretraining	Corpus connectivity, foldability, and redundancy audit	Deferred *Feasibility only*	A benchmark-safe task-specific corpus was not ready	Requires a new validated pretraining resource
Model sufficiency	Does joint substrate decoding or connected integration help?	All-A decoder, stronger-model within-pair rank gate, unified end-to-end model	Pair-first predictor, shuffled-rank control, or immutable fixed ensemble	Not supported *Outer-isolated development; unified protocol fixed before its results*	Joint decoding and connected integration are neutral or adverse in the tested comparisons	Stop tested variants; conditional repair and replication not opened
Final synthesis	Do the principles selected during development combine on unseen pairs?	Frozen two-view predictor with pair-first learning	Reproduced supplied-checkpoint AdarEdit; untouched canonical pairs	Supported held-out *Canonical held-out*	AUROC, AUPRC, F1, and Brier improve in all five reported endpoints	Validates synthesis within the benchmark construction

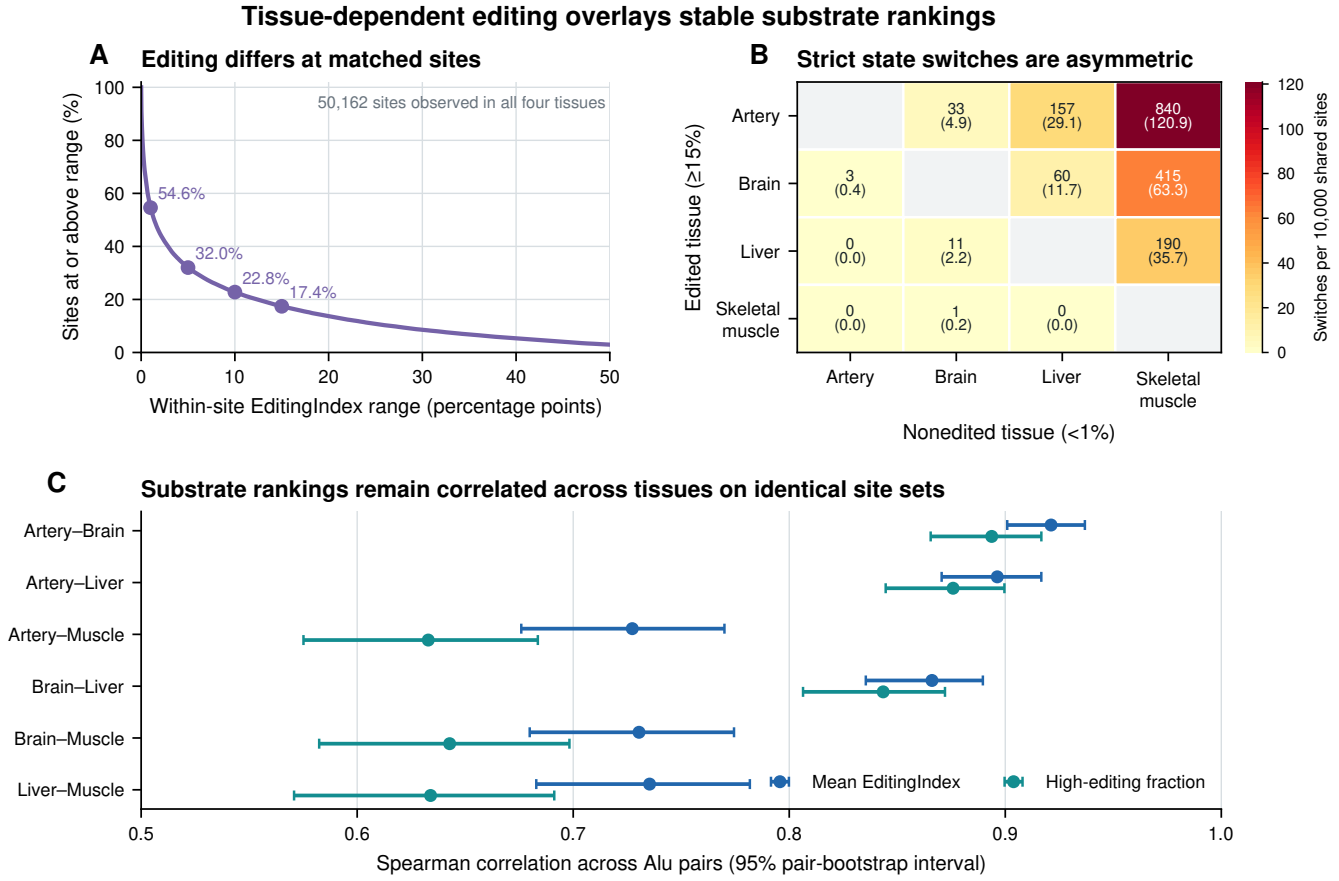


Fig. S7. Tissue-dependent editing overlays stable substrate rankings. (A) Complementary cumulative distribution of the within-site EditingIndex range for sites observed in all four individual tissues. Markers give prespecified range thresholds. (B) Strict edited/nonedited switches. Each off-diagonal cell gives the directed site count and, in parentheses, switches per 10,000 sites shared by that tissue pair; diagonal cells are undefined. (C) Spearman correlation of pair-level mean EditingIndex and high-editing-site fraction between tissues after restricting each Alu pair to identical sites and requiring at least five shared sites. Bars are 95% whole-pair bootstrap intervals. Combined is excluded throughout.

N. Post-hoc scale diagnostics

These analyses were specified after completion of the development campaign and use only retained pair-disjoint OOF predictions or matched observations from development pairs. They are descriptive tests of the multiscale interpretation, not model-selection criteria, and they do not use the canonical test.

Tissue-dependent sites and cross-tissue substrate stability. Among 74,758 sites measured in at least two individual tissues, 1,221 underwent at least one strict edited/nonedited switch. Restricting to the 50,162 sites measured in all four tissues, 54.6%, 32.0%, and 17.4% spanned at least 1, 5, and 15 EditingIndex percentage points, respectively (Fig. S7A). The directed switch matrix shows a strong tissue hierarchy but does not identify its cause (Fig. S7B).

To test substrate-level stability without coverage composition changing between tissues, each tissue pair was restricted to identical sites within each Alu pair. Pairs with at least five shared sites retained strong rank correlation in mean EditingIndex ($\rho = 0.727$–0.921) and high-editing-site fraction ($\rho = 0.633$–0.894); every pair-bootstrap interval excluded zero (Fig. S7C). Thus tissue shifts are superimposed on a reproducible substrate ranking, but neither result identifies a regulator or cellular mechanism.

Within-duplex discrimination and retained-model diagnostics. Within-duplex discrimination was evaluated only in pairs containing at least one edited and one nonedited canonical site. Each pair contributed one AUROC and received equal weight. Pair-first prediction improved mean within-pair AUROC over the internal reference in all five endpoints. In the two contexts with retained local-model OOF predictions, the fixed ensemble added +1.78 points (95% interval +1.21 to +2.40) in muscle and +1.38 points (+1.06 to +1.70) in liver. The local component alone added +1.41 points (+0.49 to +2.34) in muscle but only +0.18 points (-0.47 to +0.81) in liver. This supports ensemble-level improvement of within-substrate ranking; it does not establish an independently replicated local-component effect.

Pair-first weighting preferentially reduced log loss for substrates with fewer eligible observations, consistent with redistribution of learning influence. Descriptive Brier improvements for the local model and ensemble occurred for paired and unpaired targets and across every target-stem extent. These gains are not confined to one stratum; this analysis does not isolate which structural feature the models use.

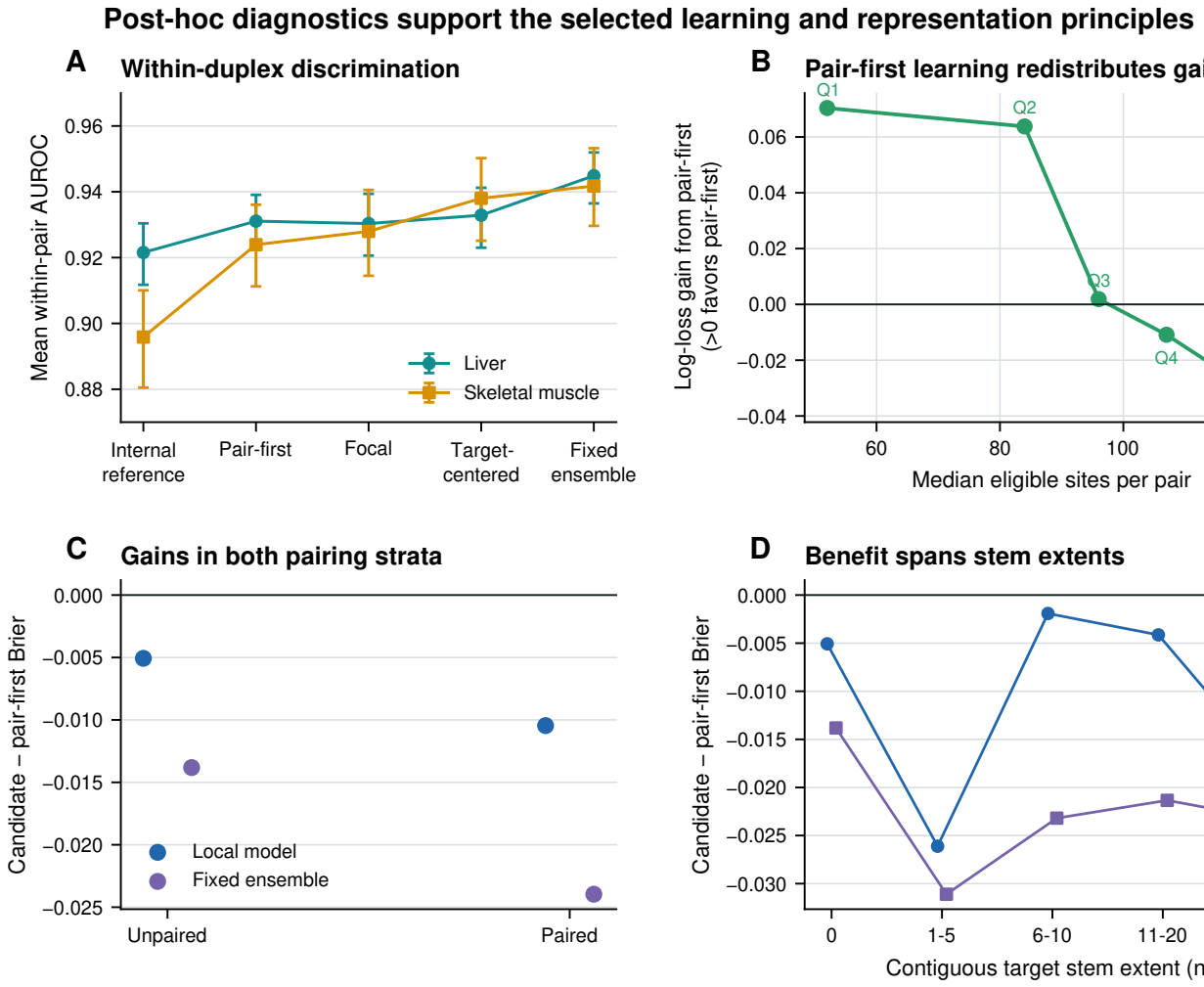


Fig. S8. Post-hoc diagnostics of selected learning and representation principles. (A) Equal-pair mean within-duplex AUROC among mixed-label pairs; bars are 95% whole-pair bootstrap intervals. The full five-endpoint pair-first comparison is in the source table. (B) Mean per-pair log-loss gain from pair-first versus row-wise expanded supervision across eligible-site-count quintiles; positive values favor pair-first. This is a descriptive development diagnostic. (C and D) Mean Brier difference relative to pair-first prediction by target paired status and contiguous target-stem extent in muscle. Negative values favor the candidate. Stratum means are descriptive and do not constitute subgroup-specific confirmatory tests.

O. Interpretation and decisive future measurements

Substrate-level propensity and target-local information leave tissue-associated differences unresolved by the tested bulk features.

A future study could pair donor- or cell-resolved editing with transcript abundance, splicing and intron retention, ADAR1 isoform and interferon state, regulator activity, quantitative chromatin, and in vivo RNA structure from the same samples. Perturbing ILF2/ILF3/ZFR-family regulators would test causal regulation separately from occupancy correlation. These are candidate experiments, not a sufficient assay set established by the present benchmark.

Artwork credits. Generic RNA motifs in main Figures 1, 2, 4, 5, and 6 were cropped, recolored, and annotated from *pri-miRNA* (designer @hakushijaco; editor Hiromasa Ono; DBCLS/TogoTV; https://doi.org/10.7875/togopic.2024.065; CC BY 4.0). They depict no sequence-specific conformation. Cell membrane and nucleus artwork in main Figures 1 and 6 was adapted from Servier Medical Art (https://smart.servier.com/), obtained through Bioicons under CC BY 3.0, recolored, and combined with RNA motifs. These compartments do not represent measured localization. Original SVGs, source revisions, licenses, and adaptation code are retained in the repository; AI-assisted composition is disclosed in the main Methods.

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